# 🌿 Algae & Photosynthesis: Guided Notes · ANSWER KEY

Biology A · Unit 2 · Level 3 (closer to the class work) · goes with Unit 2 Algae Biofuel: A1 algae + microscope, A2 algae lab, A4 Pop beads, Elodea & snails, A4.5 Stomata

Name: _____

*Use the class slides and your reading. Key words:*

*algae* · *photosynthesis* · *glucose* · *chloroplast* · *chlorophyll* · *stomata* · *guard cells* · *BTB* · *magnification*

## A. Microscopes (A1)

1. **Total magnification = eyepiece × objective**

> Example: 10x eyepiece × 10x objective = 100x

a) 10x eyepiece × 4x objective = **40x**

b) 10x eyepiece × 40x objective = **400x**

2. Always find your object on **low** power first.

## B. Photosynthesis (A4 Pop beads)

3. Photosynthesis happens in the **chloroplasts**.

4. Light energy becomes stored **chemical** energy in glucose.

5. Fill in the table.

| Inputs (go in) | Outputs (come out) |
|---|---|
| **carbon dioxide** | **glucose** |
| **water** | **oxygen** |
| **light energy** | — |

6. What happens in the light reactions?

**Light energy splits water and oxygen is released (in the thylakoids).**

## C. Stomata and BTB (A4.5, Elodea & snails, A2)

7. Guard **cells** open and close the stomata.

8. A tube has a snail and BTB in the dark. Predict the color after a day and explain.

**Yellow: the snail does respiration and adds CO2; no light means no photosynthesis to use it.**

9. In the algae lab, which tube grew the most algae and why?

**The fertilizer tube; fertilizer gives nutrients that help algae grow.**

🔑 *Key written by our team from Ms. Dunham's materials, please check.*